Control Systems || 2024-2025's MID Exam (2nd Semester):

Question No. 1 (24 points):

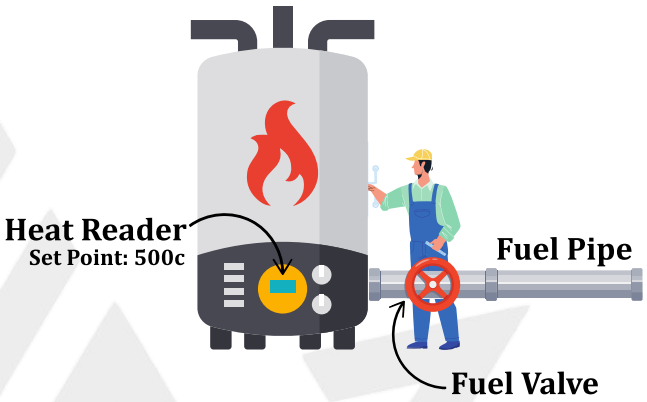
Q1. The control system below represents a:

A) Automatic Control

B) Natural Control

C) Manual Control

D) None



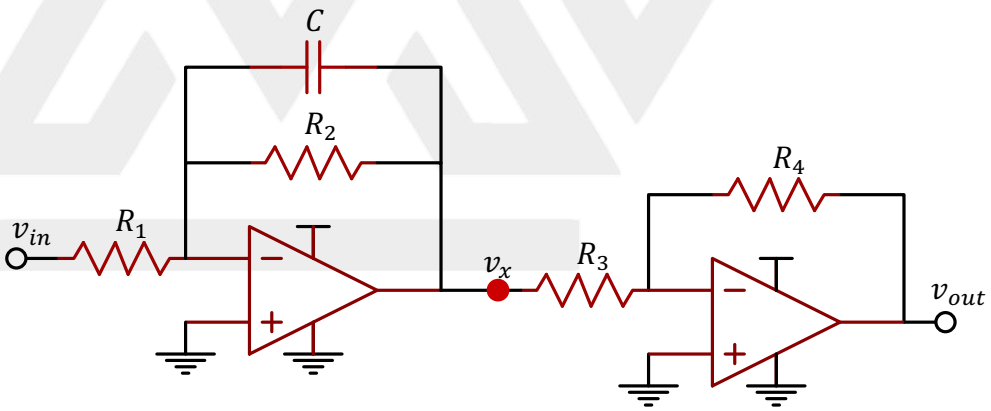
Q2. For the operational amplifier circuit below, $V_{out}/V_x =$

A) $-R_4/R_3$

B) R_4/R_3

C) $-R_4/(R_1R_2 + Cs + R_3)$

D) None



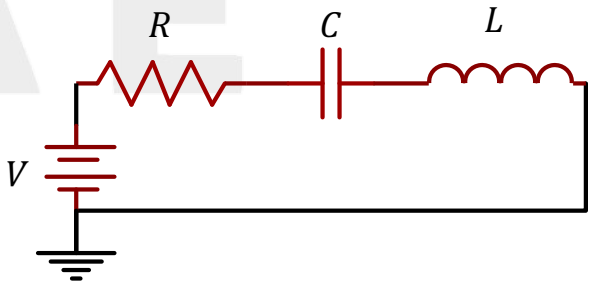
Q3. For the circuit below, the equivalent impedance $Z_T =$

A) $R + Cs + Ls$

B) $R + 1/Cs$

C) $R + Ls + 1/Cs$

D) None



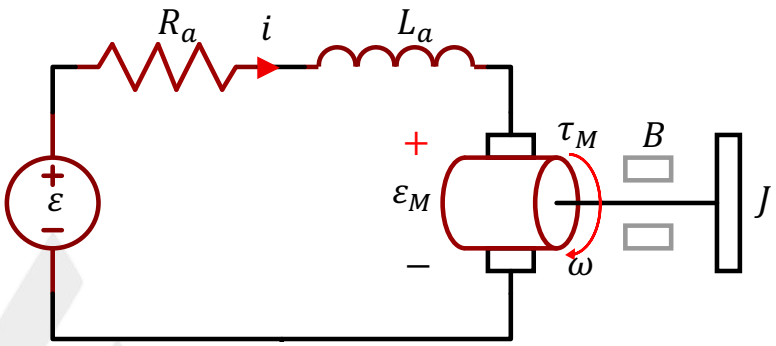
Q4. For the electromechanical system below, select the correct model:

A) $\tau_m = K_\tau i_a$, and $\varepsilon = L_a \frac{di}{dt} + R_a i_a + K_b \omega$

B) $\tau_m = K_\tau i_a$, and $\varepsilon = L_a \frac{di}{dt} + R_a i_a + K_b$

C) $\tau_m = i_a$, and $\varepsilon = L_a \frac{di}{dt} + R_a i_a + K_b$

D) None



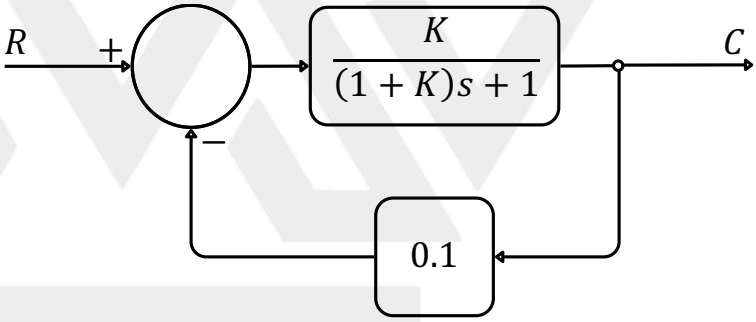
Q5. For the block diagram shown below, $E(s)/R(s) =$

A) $\frac{s+1}{(1+K)s+1+0.1K}$

B) $\frac{(1+K)s+1}{(1+K)s+1+0.1K}$

C) $\frac{(1+K)s+1}{Ks+1+0.1K}$

D) None



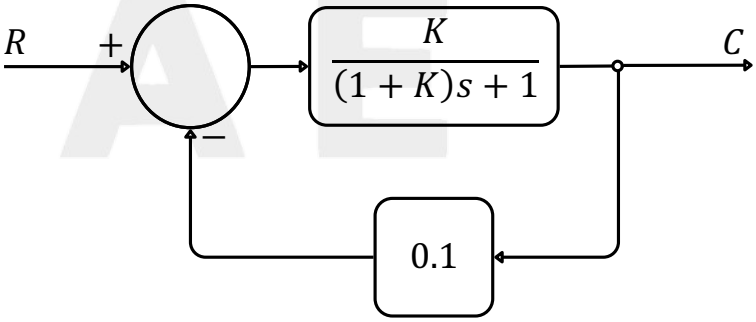
Q6. For the block diagram shown below $K = 10$. The Transfer function is:

A) $10/(11s + 0.1)$

B) $10/(11s + 2)$

C) $0.1/(11s + 1)$

D) None



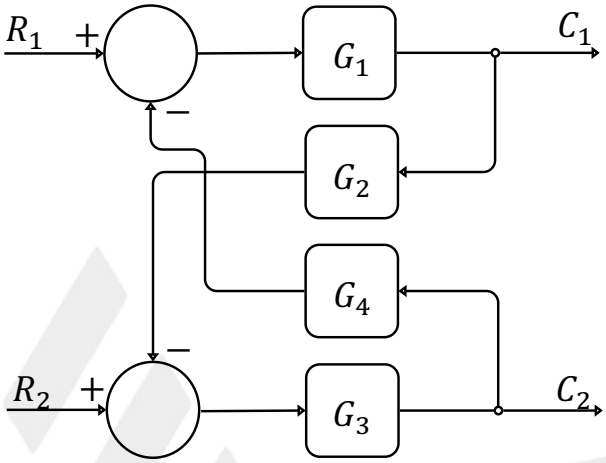
Q7. For the system below, the output C_1 due to the input R_1 is:

A) $R_1 \frac{G_1}{1-G_1G_2G_3G_4}$

B) $R_1 \frac{1}{1-G_1G_2G_3G_4}$

C) $R_1 \frac{G_1}{1+G_1G_2G_3G_4}$

D) None



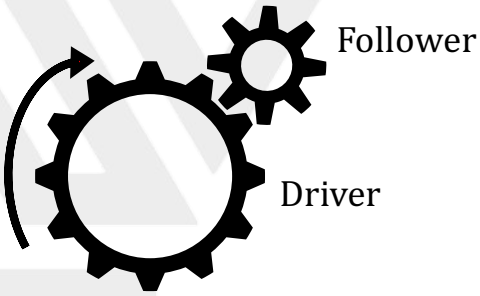
Q8. For the gear system shown below, when the driver rotates 180° the follower rotates:

A) 90°

B) 45°

C) 360°

D) None



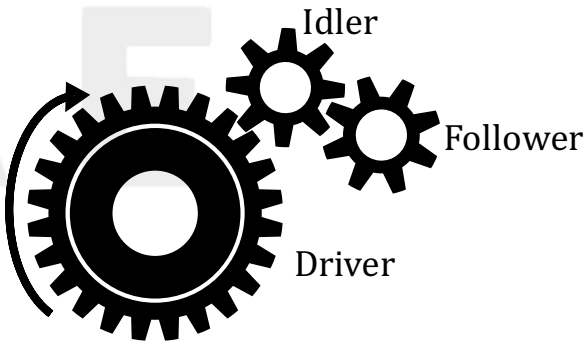
Q9. For the gear system shown below:

A) The Driver and the Idler rotate in the same direction

B) The Driver and the Follower rotate in the same direction

C) The Idler and the Follower rotate in the same direction

D) None



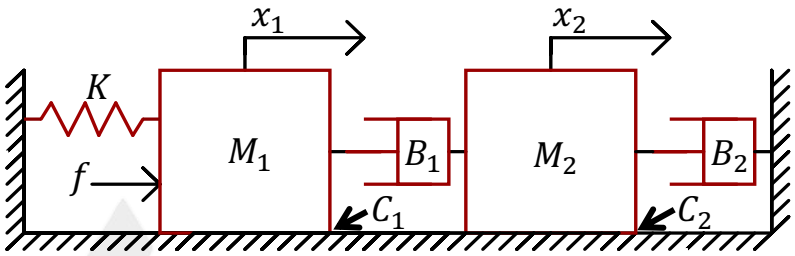
Q10. Select the equation that correctly describes the motion of the mass M_2:

A) $M_2x_2'' = B_1(x_1 - x_2) - B_2x_2' - C_2x_2'$

B) $M_2x_2'' = B_1(x_1' - x_2') - B_2x_2' - C_2x_2'$

C) $M_2x_2'' = B_1(x_1 - x_2) - B_2x_2'$

D) None



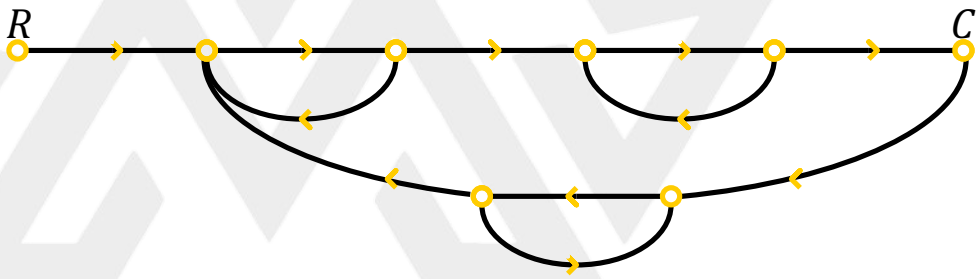
Q11. The total number of loops (touching and non-touching) in the signal flow graph below is:

A) One loop

B) Two loops

C) Three loops

D) Four loops



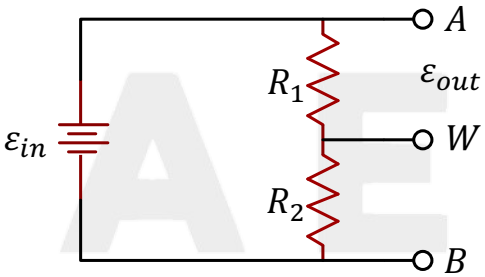
Q12. Following are equations representing the potentiometer output voltage ε_{out}, the wrong equation is:

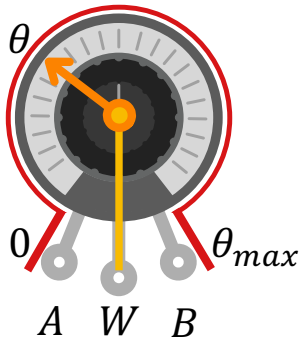
A) $\varepsilon_{out} = \varepsilon_{in} \frac{R_2}{R_1+R_2}$

B) $\varepsilon_{out} = \varepsilon_{in} \frac{\theta}{\theta_{max}}$

C) $\varepsilon_{out} = \varepsilon_{in} \frac{R_1}{R_1+R_2}$

D) None

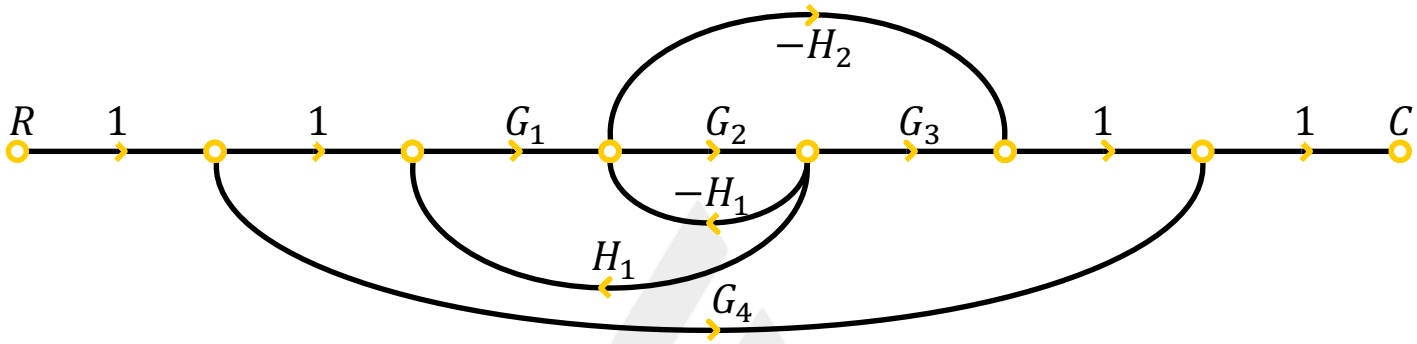




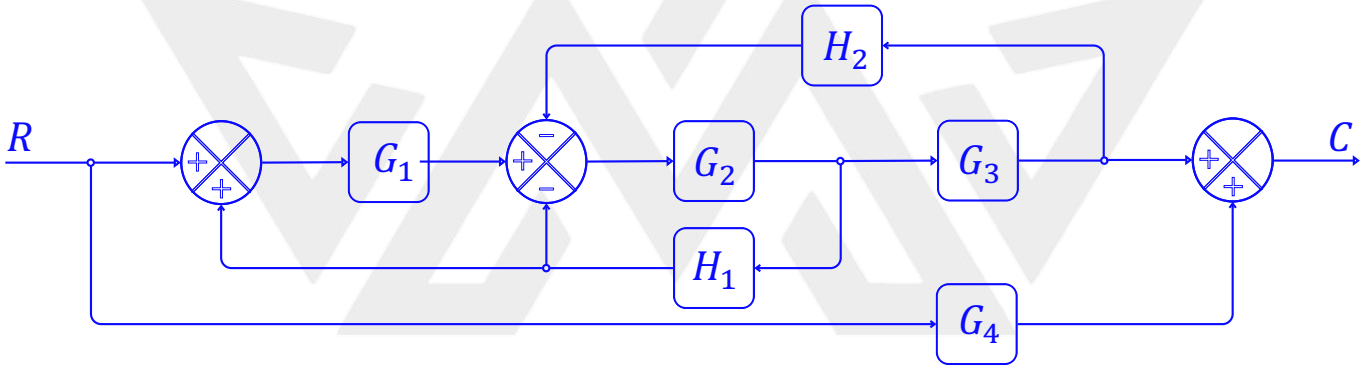
Q	1	2	3	4	5	6	7	8	9	10	11	12
Ans	C	A	C	A	B	B	A	C	B	B	D	A

Question No. 2 (6 points):

Represent the following Signal Flow Graph as a Block Diagram and give the forward paths.



Block Diagram:



Forward Paths:

$$P = G_1G_2G_3, \qquad P_2 = G_4$$

Created By Mohammad Abu Elyan.